

KHAOULA TRABELSI

Master's Student in Automatic Control, Robotics & Information Processing

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Portfolio

Profile

Master's student in Automatation, Robotics and Information Processing, with a multidisciplinary background in robotics, automatic control, artificial intelligence, signal processing and embedded systems. Experience in robot kinematics and simulation, machine learning and deep learning, IoT data acquisition, embedded software and engineering simulation. Interested in research projects related to **AI for Robotics, intelligent control, autonomous systems, robot manipulation and computer vision**.

Education

École Nationale d'Ingénieurs de Carthage (ENICarthage)	2025 – Present
Master of Research in Automatation, Robotics and Information Processing	Charguia, Tunisia
Institut Supérieur des Technologies de l'Information et de la Communication (ISTIC)	2021 – 2024
Bachelor's Degree in Electronics, Electrical Engineering and Automatic Control	
Specialization: Automatic Control and Industrial Computing	Borj Cedria, Tunisia
Lycée Cité El Salem	2017 – 2021
Baccalaureate in Mathematics	Boumhal, Ben Arous

Research Interests

Robotics: Robot Kinematics, Robot Manipulation, Autonomous Systems
Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision
Control: Automatic Control, Intelligent Systems, Robot Modeling
Information Processing: Signal Processing, Image Processing, Wireless Communications

Professional Experience

PFE Internship	January 2024 – May 2024
<i>TIS Circuits, Ben Arous, Tunisia</i>	
- Designed and developed a complete IoT data acquisition board for measurement and calibration applications. - Developed a Python/Tkinter metrology platform for calibration data visualization and analysis. - Automated calibration procedures and generated detailed measurement reports. - Worked with data acquisition and communication protocols.	
PFA Internship	July 2023 – September 2023
<i>Sagemcom Software & Technology, Megrine, Tunisia</i>	
- Rewrote software testing tools from Shell to Python. - Designed the software architecture of the new testing tools. - Performed testing and validation on electronic boards. - Worked in an embedded Linux environment and with TFTP-based tools.	

Technical Skills

Robotics & Control: Robot Kinematics, DH Modeling, Homogeneous Transformations, Jacobian Analysis, MATLAB, Simulink
Artificial Intelligence: Machine Learning, Scikit-learn, KNN, SVM, Logistic Regression, Deep Learning, CNN, TensorFlow, Keras, Image Classification
Signal Processing & Communications: FFT, OFDM, BER/SNR Analysis, LDPC/Turbo Codes, PAPR Reduction, Wi-Fi 6/7, 5G
Programming: Python, C++, VHDL, JavaScript, HTML/CSS
Software & Development: Git, ReactJS, Node.js, MongoDB, Tkinter
Embedded & IoT: Microcontrollers, IoT Data Acquisition, Communication Protocols, Embedded Linux
Engineering Simulation: LTspice, CST Studio Suite, FDTD, Altium Designer, SPICE

Selected Academic Projects

Geometric Modeling of Robotic Manipulators	2025
<i>Master Project – ENICarthage</i>	
- Established coordinate frames and extracted Denavit–Hartenberg parameters from robot kinematic structures. - Derived homogeneous transformation matrices and the complete forward kinematic model. - Implemented numerical modeling and 3D simulation using MATLAB. - Analyzed the Jacobian matrix for velocity and singularity analysis.	

Image Classification using Machine Learning and Deep Learning **2025**

Academic Project

- Prepared and analyzed an image dataset for supervised learning.
- Implemented and compared KNN, Logistic Regression and SVM classification models using Scikit-learn.
- Developed a CNN-based image classification model using TensorFlow/Keras.
- Compared model performance and analyzed overfitting and generalization behavior.

OFDM System Implementation and Optimization **2025**

Academic Project

- Implemented and analyzed a complete OFDM communication chain using MATLAB/Simulink.
- Studied LDPC/Turbo coding, PAPR reduction and BER/SNR performance.
- Analyzed the impact of system parameters on communication performance.

EMI Analysis of a Buck Converter **2025**

Master Project – ENICarthage

- Modeled a Buck converter including parasitic elements using SPICE.
- Performed parametric analysis of input voltage, switching frequency and load current.
- Automatically extracted FFT spectra for differential-mode and common-mode emissions.
- Developed a Python GUI for visualization and analysis of results.

Double-Band Dual-Mode Bandpass Filter Design

CST Studio Suite – Academic Project

- Simulated an Open Loop Resonator (OLR) with stub using CST.
- Designed the geometry and validated simulation results against published reference data.

Languages

Arabic: Native | **French:** B1 | **English:** B2

Academic & Professional Activities

IEEE – Institute of Electrical and Electronics Engineers

Active Member

Member of the *IEEE Computer Society Chapter*.

Soft Skills

Analytical Thinking, Research, Problem Solving, Adaptability, Teamwork, Autonomy, Rigor, Intellectual Curiosity